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07/10 :-

Kubernetes. vs. Docker Swarm.
↳ UI Support ↳ Command lines.
↳ Monitor Support ↳ Only metrics.

Kubernetes files → resource object.

(*) Docker engine should be running.

Minikube :-
↳ It does not have clusters.
↳ Works with single node (Master node).

→ minikube version
To check version.

→ minikube start
To start minikube.

→ minikube kubectl --get nodes
Used to list the clusters & its nodes.

→ minikube kubectl -get pods.
List the pods that are created.

→ minikube kubectl -get pods -A
List all the component which defaultly assigns in a pod, it will show those components.

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→ minikube kubectl -- run apachepod
--image - httpd.
↳ To create a pod. pod name.

→ minikube kubectl - describe pod apachepod.
↳ Gives metadata of the pod.
↓
(Data Abt data).

→ minikube kubectl --logs apachepod
↳ Application status in the container which inside pod, give reports abt the status

→ minikube kubectl -- exec -it apachepod
-- /bin/bash
↳ Execute the pod.

To update :-
root@> apt update.

→ minikube kubectl -- delete pod apachepod
↳ To delete pod.

Kubernetes can be declared / accessed by 2 methods
1) Declarative method. (yaml file)
2) Imperative method. (commands).

kubernetes.io → documentation website

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→ minikube kubectl -- apply -f simplepod.yaml
↳ Use for resource object (kubernetes file).

→ minikube kubectl -- delete -f simplepod.yaml

deploy
↳ load Balancing, Self-healing, auto-update, rolling update.

Rolling deployment (Rolling Update).
↳ Undoing implementation = rollback.
↳ Changing containers after changed container is stabilized.

→ minikube kubectl -- create deployment demodeploy --image=nginx:1.36.
↓ ↓
deployment name image name.

→ minikube kubectl -- get deploy
↳ list the deployments

→ minikube kubectl -- scale deployment demodeploy --replicas=5
↳ Will create 5 replicas of deployment.

→ minikube kubectl -- get nginx -o.
↓
pod name.

Rolling Updates :-

→ minikube kubectl -- set image
deployment/nginxdeploy
↳ updates image version from
nginx = nginx : 1.30.5.

→ minikube kubectl -- set image deployment/
nginxdeploy httpd
↳ update the whole image from nginx
to httpd.

Auto healing → delete all pods and check for
pods it will create again.
↳ Delete all nodes and describe the nodes
it will create a new whole pods.

→ minikube kubectl -- rollout status
deployment nginxdeploy.
↳ Set image kudukura appo, rolling
status check panrathuku.

→ minikube kubectl -- rollout set image
deployment/nginxdeploy nginx=1:30:5=node.
↓ ↓
current image / dinre image

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→ minikube kubectl -rollout undo deployment nginxdeploy.

↳ previous version of image retrieve panna undo.

Networks:

Service → To have a static IP. called service.ip (SIP)

To have static IP - service IP

↳ DNS.

Kube proxy.

↳ It will redirect the IP from SIP to Pods.

Node to Node Communication:

↳ Route to route communication.

↳ Route table.

: CNI - 10.

↳ Default by kubernetes